

Characteristics of restricted selection indices and geometrical interpretation of restricted breeding values

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Abstract

Restricted selection is used to control genetic changes in one or more characters. Three main selection indices are adopted for this purpose. First, Kempthorne's index is used to maximize aggregate breeding value (BV) with changes in some traits restricted to zero; second, Harville's index is used to maximize aggregate BV with proportional changes for some traits; and third, Yamada's index is mathematically used to achieve the relative desired changes for all traits. Kempthorne's index is equivalent to Harville's index. However, the relationship between Kempthorne's and Yamada's indices has not been clarified. In addition, the characteristics of restricted selection indices and the relationship between BV and restricted BV (RBV) are also unknown. The aim of this study was to clarify the characteristics of restricted selection indices and describe the relationship between BV and RBV by using linear algebra and geometric techniques, respectively. First, I proved that Yamada's index is part of Kempthorne's index. Second, I investigated the relationship between BVs that were estimated using an ordinary selection index (EBVs) and RBVs estimated using a restricted selection index (ERBVs) and proved that the ERBVs of the restricted traits are proportional to the relative desired changes. Third, I proved that RBV is represented by a linear function of BV and geometrically represented the relationship between BV and RBV. In this study, new findings on restricted selection indices and RBV were obtained. This useful clarification of the relationship between BV and RBV will make it possible to evaluate the response to selection using not only a restricted selection index, but also a restricted BLUP in computer simulation studies.

KEYWORDS

proportional changes, relative desired changes, restricted breeding values, restricted selection index, zero changes

1 | INTRODUCTION

A linear selection index for estimating aggregate breeding values (BVs) was originally introduced in animal breeding by Hazel (1943). Subsequently, several selection indices

have been introduced, such as restricted selection (e.g., Harville, 1975; Kempthorne & Nordskog, 1959; Yamada et al., 1975), multi-stage selection (e.g., Cunningham, 1975; Xu & Muir, 1992; Young, 1964), selection for nonlinear profit function (e.g., Goddard, 1983; Itoh & Yamada, 1988; Wilton

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et al., 1968), and molecular eigen selection (e.g., Cerón-Rojas et al., 2008a, 2008b) indices. To date, selection index theory has been used to estimate the accuracy of selection and the expected genetic gains by BLUP and genomic selection (e.g., Barwick et al., 2013; Bijma & Dekkers, 2022; Dekkers, 2007).

Restricted selection is used to control genetic changes in one or more characters for improvement and is a particularly efficient method for antagonistic selection. Kempthorne and Nordskog (1959) originally proposed the concept of imposing restrictions on conventional selection indices. Their selection index maximizes the aggregate BV, with changes in some traits restricted to zero. Harville (1975) and Tallis (1985) derived different selection indices to maximize aggregate BV with proportional changes for some traits. Yamada et al. (1975) proposed a selection index to achieve the relative desired changes for all traits.

Among these practical selection indices, Itoh and Yamada (1987) proved that the three indices derived by Kempthorne and Nordskog (1959), Harville (1975), and Tallis (1985) are mathematically equivalent. However, the relationship between Kempthorne's and Yamada's selection indices has not been clarified for nearly half a century. Hence, the first objective of this study was to determine the relationship between them.

In general, as the ordinary selection index (Hazel, 1943) estimates the aggregate BV directly from the phenotypic value for each trait, the BV for each trait is rarely considered. Consequently, the relationship between the BVs estimated by the ordinary selection index (EBV) and restricted BV (RBV) estimated by the restricted selection index (ERBV) is not obvious. In particular, the concept of BV estimation does not exist in the selection index for achieving the relative desired changes. The second objective of this study was to clarify the characteristics of the RBVs estimated using a restricted selection index.

As the true BV is already known in a selection experiment using a Monte Carlo computer simulation, the response to selection can be evaluated from the BV for each trait and the aggregate BV. However, in the response to restricted selection, such as that obtained from using restricted selection index or restricted BLUP methods, evaluation criteria for response to selection do not exist. The third objective of this study was to clarify the relationship between BV and RBV. As an example of this application, a computer simulation was used to evaluate the response to selection.

2 | THEORY

2.1 | The index of Kempthorne and Nordskog (1959)

The aim of this selection index is to maximize the aggregate BV, with changes in some traits restricted to

zero, in a population. To achieve this, the square of the correlation coefficient between the selection index ($I = \mathbf{b}'\mathbf{p}$) and aggregate BV ($H = \mathbf{a}'\mathbf{g}$) is maximized under $\mathbf{C}'\mathbf{G}'\mathbf{b} = \mathbf{0}$ and $\mathbf{b}'\mathbf{P}\mathbf{b} = 1$, where $\mathbf{b}$ is the vector of coefficients of the selection index, $\mathbf{p}$ is the vector of phenotypic values, $\mathbf{a}$ is the vector of economic weights, $\mathbf{g}$ is the vector of BVs, $\mathbf{C}$ is the matrix for restrictions (Itoh & Yamada, 1987; Mallard, 1972; Satoh, 2004), $\mathbf{G} = \text{cov}(\mathbf{p}, \mathbf{g}')$, and $\mathbf{P} = \text{var}(\mathbf{p})$. The coefficient vector ($\mathbf{b}_K$) of the selection index that satisfies these conditions is calculated as follows:

$$\mathbf{b}_K = [\mathbf{I} - \mathbf{P}^{-1}\mathbf{G}\mathbf{C}(\mathbf{C}'\mathbf{G}'\mathbf{P}^{-1}\mathbf{G}\mathbf{C})^{-1}\mathbf{C}'\mathbf{G}']\mathbf{P}^{-1}\mathbf{G}\mathbf{a} \quad (1)$$

where $\mathbf{I}$ denotes an identity matrix. Itoh and Yamada (1987) proved that this index is equivalent to those derived by Harville (1975) and Tallis (1985). Hence, the selection index with changes in some traits restricted to zero is equivalent to that with proportional changes for some traits.

2.2 | The index of Yamada et al. (1975)

The aim of this selection index is to achieve the relative desired changes for all traits. Yamada et al. (1975) derived this index by using the relationship between the genetic response to selection ($\Delta\mathbf{g}$) and the relative desired changes ($\mathbf{k}$). The coefficient vector ($\mathbf{b}_Y$) of this index is expressed as follows:

$$\mathbf{b}_Y = (\mathbf{G}')^{-1}\mathbf{k}. \quad (2)$$

If the numbers of the selected and objective traits are different, the coefficient vector is calculated as follows:

$$\mathbf{b}_Y = \mathbf{P}^{-1}\mathbf{G}(\mathbf{G}'\mathbf{P}^{-1}\mathbf{G})^{-1}\mathbf{k}.$$

2.3 | Computer simulation

To evaluate the response to selection with constraint, a Monte Carlo computer simulation was used to generate pedigree and phenotypic records. The base population (G0) consisted of ten males and 50 or 100 females that were all assumed to be unrelated, unselected, and non-inbred. Three traits of six separate generations (G0 to G5) were simulated. Each male was randomly mated with an equal number of females (5 or 10 females). A mated female produced two males and two females, and ten males and 50 or 100 females were randomly selected in each generation. Animals in G0 had no record, and G1 through G5 had one record in each trait for all animals. An infinitesimal additive genetic model (Bulmer, 1980) was assumed. The following genetic parameters were used:

$$\mathbf{P} = \begin{bmatrix} 1.0 & 0.25 & 0.25 \\ 0.25 & 1.0 & 0.25 \\ 0.25 & 0.25 & 1.0 \end{bmatrix}, \mathbf{G} = \begin{bmatrix} 0.2 & 0.1 & 0.1 \\ 0.1 & 0.3 & 0.1 \\ 0.1 & 0.1 & 0.4 \end{bmatrix}, \text{ and } \mathbf{k} = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}.$$

Two selection methods were used: the restricted selection index and the restricted BLUP. BVs were estimated using the genetic parameters that generated the data or the falsely estimated 0.1 instead of the additive genetic variance of 0.2 for the first trait. One hundred replicates were simulated for each combination ($2 \times 2 \times 2 = 8$) of selection method, genetic parameter predicted restricted BV, and population size. The deviation of the average BV of animals selected in G5 from the population mean of BV in G5 was used as the response to selection.

3 | RESULTS

3.1 | Equivalence of restricted selection indices

The following lemma was used to prove the equivalence of the selection indices of Kempthorne and Nordskog (1959) and Yamada et al. (1975); Appendix 1 presents the proof of this lemma.

Lemma 1. Let $\mathbf{X}_{n \times r}$, $\mathbf{Y}_{n \times (n-r)}$ be of rank r and $(n-r)$ such that $\mathbf{Y}'\mathbf{X} = \mathbf{0}$. If $\mathbf{M}_{n \times n}$ is a symmetric positive definite matrix, then

$$\mathbf{Y}(\mathbf{Y}'\mathbf{M}\mathbf{Y})^{-1}\mathbf{Y}' = \mathbf{M}^{-1} - \mathbf{M}^{-1}\mathbf{X}(\mathbf{X}'\mathbf{M}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{M}^{-1}.$$

Let n be the number of selected traits and r be the number of restrictions. From Lemma 1,

$$\mathbf{k}(\mathbf{k}'(\mathbf{G}'\mathbf{P}^{-1}\mathbf{G})^{-1}\mathbf{k})^{-1}\mathbf{k}' = \mathbf{G}'[\mathbf{I} - \mathbf{P}^{-1}\mathbf{G}\mathbf{C}(\mathbf{C}'\mathbf{G}'\mathbf{P}^{-1}\mathbf{G}\mathbf{C})^{-1}\mathbf{C}'\mathbf{G}']\mathbf{P}^{-1}\mathbf{G}.$$

Multiplying $\mathbf{G}'$ from the left side of Equation (1),

$$\begin{aligned} \mathbf{G}'\mathbf{b}_K &= \mathbf{G}'[\mathbf{I} - \mathbf{P}^{-1}\mathbf{G}\mathbf{C}(\mathbf{C}'\mathbf{G}'\mathbf{P}^{-1}\mathbf{G}\mathbf{C})^{-1}\mathbf{C}'\mathbf{G}']\mathbf{P}^{-1} \\ \mathbf{G}\mathbf{a} &= \mathbf{k}(\mathbf{k}'(\mathbf{G}'\mathbf{P}^{-1}\mathbf{G})^{-1}\mathbf{k})^{-1}\mathbf{k}'\mathbf{a} = \alpha\mathbf{k}, \end{aligned}$$

where α is a scalar and $\alpha = (\mathbf{k}'(\mathbf{G}'\mathbf{P}^{-1}\mathbf{G})^{-1}\mathbf{k})^{-1}\mathbf{k}'\mathbf{a}$. As α is merely a constant multiplier for $\mathbf{k}$,

$$\mathbf{G}'\mathbf{b}_K = \mathbf{k}.$$

This $\mathbf{b}_K$ is the same as $\mathbf{b}_Y$, which is the coefficient vector of Yamada's index (Equation 2). On the other hand, if the number of rows in matrix $\mathbf{G}$ is greater than the number of columns, Kempthorne's index cannot be derived from Yamada's index. Therefore, the index of

Yamada et al. (1975) is proved to be considered as part of Kempthorne and Nordskog (1959).

3.2 | Characteristics of BVs estimated by restricted selection index

First, I clarify the relationship between the BVs estimated using an ordinary selection index (EBVs: $\hat{\mathbf{g}}$) and those estimated by the restricted selection index (ERBVs: $\hat{\mathbf{g}}_R$). The ordinary selection index (I_H) is given as follows:

$$I_H = \mathbf{b}'_H\mathbf{p} = \mathbf{a}'\hat{\mathbf{g}} = \mathbf{a}'\mathbf{G}'\mathbf{P}^{-1}\mathbf{p}$$

where $\mathbf{b}_H$ is the coefficient vector of I_H (Hazel, 1943). From Equation (1), the restricted selection index (I_R) is given as follows:

$$I_R = \mathbf{b}'_R\mathbf{p} = \mathbf{a}'\hat{\mathbf{g}}_R = \mathbf{a}'\mathbf{G}'\mathbf{P}^{-1}[\mathbf{I} - \mathbf{G}\mathbf{C}(\mathbf{C}'\mathbf{G}'\mathbf{P}^{-1}\mathbf{G}\mathbf{C})^{-1}\mathbf{C}'\mathbf{G}'\mathbf{P}^{-1}]\mathbf{p}$$

where $\mathbf{b}_R$ is the coefficient vector of I_R . Hence,

$$\hat{\mathbf{g}}_R = [\mathbf{I} - \mathbf{G}'\mathbf{P}^{-1}\mathbf{G}\mathbf{C}(\mathbf{C}'\mathbf{G}'\mathbf{P}^{-1}\mathbf{G}\mathbf{C})^{-1}\mathbf{C}']\hat{\mathbf{g}}. \quad (3)$$

Next, I proved that the ERBVs of the restricted traits are proportional to the relative desired changes ($\mathbf{k}_0$). Let $\mathbf{g}_1$ and $\mathbf{g}_2$ be the BVs for traits with and without constraints, respectively, $\mathbf{G}_1 = \text{cov}(\mathbf{p}, \mathbf{g}_1)$, and $\mathbf{G}_2 = \text{cov}(\mathbf{p}, \mathbf{g}_2)$. The restriction matrix $\mathbf{C}$ corresponding to $\mathbf{g}$ can be written as $\mathbf{C} = [\mathbf{C}'_0 \ \mathbf{0}]'$, where $\mathbf{C}'_0\mathbf{k}_0 = \mathbf{0}$. Then, from $\mathbf{G}\mathbf{C} = \mathbf{G}_1\mathbf{C}_0$,

where

$$\begin{aligned} \hat{\mathbf{g}}_R &= \mathbf{G}'\mathbf{P}^{-1}[\mathbf{I} - \mathbf{G}\mathbf{C}(\mathbf{C}'\mathbf{G}'\mathbf{P}^{-1}\mathbf{G}\mathbf{C})^{-1}\mathbf{C}'\mathbf{G}'\mathbf{P}^{-1}]\mathbf{p} \\ &= \begin{bmatrix} \hat{\beta}_0\mathbf{k}_0 \\ [\mathbf{G}'_2 - \mathbf{P}^{-1}\mathbf{G}_1\mathbf{C}_0(\mathbf{C}'_0\mathbf{G}'_1\mathbf{P}^{-1}\mathbf{G}_1\mathbf{C}_0)^{-1}\mathbf{C}'_0\mathbf{G}'_1]\mathbf{P}^{-1}\mathbf{p} \end{bmatrix} \end{aligned}$$

$$\hat{\beta}_0 = [\mathbf{k}'_0(\mathbf{G}'_1\mathbf{P}^{-1}\mathbf{G}_1)^{-1}\mathbf{k}_0]^{-1}\mathbf{k}'_0\mathbf{P}^{-1}\mathbf{G}_1(\mathbf{G}'_1\mathbf{P}^{-1}\mathbf{G}_1)^{-1}\mathbf{p}$$

(Appendix 2). Similarly, when restrictions are applied for all traits,

$$\hat{\mathbf{g}}_R = \mathbf{k}[\mathbf{k}'(\mathbf{G}'\mathbf{P}^{-1}\mathbf{G})^{-1}\mathbf{k}]^{-1}\mathbf{k}'\mathbf{P}^{-1}\mathbf{G}(\mathbf{G}'\mathbf{P}^{-1}\mathbf{G})^{-1}\mathbf{p} = \hat{\beta}_1\mathbf{k}$$

where $\hat{\beta}_1 = [\mathbf{k}'(\mathbf{G}'\mathbf{P}^{-1}\mathbf{G})^{-1}\mathbf{k}]^{-1}\mathbf{k}'(\mathbf{G}'\mathbf{P}^{-1}\mathbf{G})^{-1}\mathbf{G}'\mathbf{P}^{-1}\mathbf{p}$. Therefore, the ERBVs of restricted traits were proportional to the relative desired changes.

Furthermore, the estimate of restricted aggregate breeding value ($\hat{H}_R$) is given as follows:

$$\hat{H}_R = \mathbf{a}'\hat{\mathbf{g}}_R = \hat{\beta}_1\mathbf{a}'\mathbf{k}.$$

Since $\mathbf{a}'\mathbf{k}$ is a constant, $\hat{H}_R$ is independent of the economic weights and only depends on the magnitude of $\hat{\beta}_1$. In other words, the magnitude of $\hat{\beta}_1$ is the selection criterion when using the selection for relative desired changes.

3.3 | Geometrical interpretation of restricted BVs

If the random variables BV in $\mathbf{g}$ have an n -dimensional normal distribution with a vector of mean $\mathbf{0}$ and variance-covariance matrix $\mathbf{G}_0$, the joint density function is given as follows:

$$F(\mathbf{g}) = \frac{1}{2\pi |\mathbf{G}_0|^{\frac{1}{2}}} \exp\left(-\frac{1}{2} \mathbf{g}' \mathbf{G}_0^{-1} \mathbf{g}\right).$$

When $\mathbf{g}' \mathbf{G}_0^{-1} \mathbf{g}$ is a constant ($= \gamma > 0$), the joint probability hyperelliptical equation of the BV is given as follows:

$$\mathbf{g}' \mathbf{G}_0^{-1} \mathbf{g} - \gamma = 0 \quad (4)$$

The equation of the tangent hyperplane at point $\mathbf{g}_T$ on the hypersurface (4) is given as follows:

$$\mathbf{g}'_T \mathbf{G}_0^{-1} \mathbf{g} = \mathbf{g}'_T \mathbf{G}_0^{-1} \mathbf{g}_T. \quad (5)$$

Therefore, $\mathbf{g}'_T \mathbf{G}_0^{-1}$ is the normal vector of the tangent hyperplane. From $\mathbf{g}'_T \mathbf{G}_0^{-1} \mathbf{G}_0 \mathbf{C} = \mathbf{0}$, a set of points through the origin and parallel to (5) is the $(n-r)$ -dimensional transformation space:

$$\mathbf{g}_T = \mathbf{G}_0 \mathbf{C} \mathbf{t}$$

where $\mathbf{t}$ is any scalar when $r=1$, otherwise any r -dimensional vector and $\text{rank}(\mathbf{C})=r$. For example, the set of points $\mathbf{g}_T$ is a plane when $r=2$. Hence, the vector space that passes through point $\mathbf{g}_{R_0}$ on the restricted hyperplane and parallels $\mathbf{g}_T = \mathbf{G}_0 \mathbf{C} \mathbf{t}$ is represented as follows:

$$\mathbf{g} = \mathbf{g}_{R_0} + \mathbf{G}_0 \mathbf{C} \mathbf{t}. \quad (6)$$

Let any point on (6) be $\mathbf{g}_0$. As $\mathbf{t} = (\mathbf{C}' \mathbf{G}_0 \mathbf{C})^{-1} \mathbf{C}' (\mathbf{g}_0 - \mathbf{g}_{R_0})$, then

$$\mathbf{g}_0 = \mathbf{g}_{R_0} + \mathbf{G}_0 \mathbf{C} (\mathbf{C}' \mathbf{G}_0 \mathbf{C})^{-1} \mathbf{C}' (\mathbf{g}_0 - \mathbf{g}_{R_0}).$$

Using $\mathbf{C}' \mathbf{g}_{R_0} = \mathbf{0}$,

$$\mathbf{g}_{R_0} = [\mathbf{I} - \mathbf{G}_0 \mathbf{C} (\mathbf{C}' \mathbf{G}_0 \mathbf{C})^{-1} \mathbf{C}'] \mathbf{g}_0. \quad (7)$$

Therefore, the set of RBV ($\mathbf{g}_{R_0}$) is geometrically on the $(n-r)$ -dimensional restricted hyperplane, $\mathbf{C}' \mathbf{g} = \mathbf{0}$, and the RBV represented by $\mathbf{g}_0$ is the point of intersection of the restricted hyperplane and the r -dimensional vector space that parallels $\mathbf{g} = \mathbf{G}_0 \mathbf{C} \mathbf{t}$ and passes through $\mathbf{g}_0$.

A restriction regarding predetermined relative changes for all traits is a special case within the category of constraints, with changes in some traits being restricted to zero. Suppose that the vector of the line of proportional changes is $\mathbf{g} = \beta \mathbf{k}$, where β is any scalar. This vector agrees with the restricted hyperplane, $\mathbf{C}' \mathbf{g} = \mathbf{0}$. That is, the intersection of $\mathbf{g} = \beta \mathbf{k}$ and hyperellipsoid (4) corresponds to point $\mathbf{g}_{R_0}$ on the restricted space. Let $\mathbf{g}_0$ be any point on tangent hyperplane (5) at point $\mathbf{g}_{R_0}$,

$$\mathbf{g}_{R_0} = \beta \mathbf{k} \quad (8)$$

and

$$\mathbf{g}'_{R_0} \mathbf{G}_0^{-1} \mathbf{g}_0 = \mathbf{g}'_{R_0} \mathbf{G}_0^{-1} \mathbf{g}_{R_0}. \quad (9)$$

From Equations (8) and (9),

$$\mathbf{g}_{R_0} = \mathbf{k} (\mathbf{k}' \mathbf{G}_0^{-1} \mathbf{k})^{-1} \mathbf{k}' \mathbf{G}_0^{-1} \mathbf{g}_0. \quad (10)$$

When a restriction consists of predetermined relative changes for all traits, $\text{rank}(\mathbf{C}) = n-1$. From Lemma 1,

$$\mathbf{k} (\mathbf{k}' \mathbf{G}_0^{-1} \mathbf{k})^{-1} \mathbf{k}' = \mathbf{G}_0 - \mathbf{G}_0 \mathbf{C} (\mathbf{C}' \mathbf{G}_0 \mathbf{C})^{-1} \mathbf{C}' \mathbf{G}_0.$$

As a result, Equation (10) can be expressed as follows:

$$\mathbf{g}_{R_0} = [\mathbf{I} - \mathbf{G}_0 \mathbf{C} (\mathbf{C}' \mathbf{G}_0 \mathbf{C})^{-1} \mathbf{C}'] \mathbf{g}_0.$$

This implies that the restricted space is a straight line of proportional changes, and the RBV of $\mathbf{g}_0$ is on the straight line. In this case, therefore, the RBV from the selection index of Kempthorne (Equation 7) is in agreement with that of Yamada (Equation 10). The relationship between BV and RBV is given as follows:

$$\mathbf{g}_R = [\mathbf{I} - \mathbf{G}_0 \mathbf{C} (\mathbf{C}' \mathbf{G}_0 \mathbf{C})^{-1} \mathbf{C}'] \mathbf{g} = \beta \mathbf{k}. \quad (11)$$

The β represents the genetic evaluation criterion for animals when using the selection for relative desired changes. This criterion with known BV such as computer simulations can be used not only for the restricted

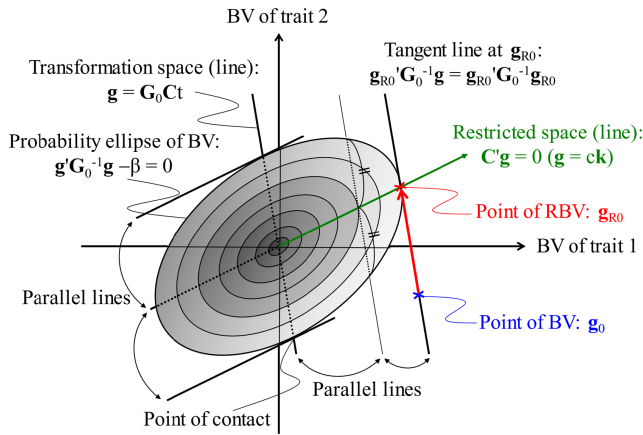


FIGURE 1 The relationship between BV and RBV in a restricted selection of two traits for improving trait 1 and no-changing trait 2. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/jbg.12845)]

selection index method, but also for all restricted selection methods, including the restricted BLUP method (Quaas & Henderson, 1976) and the method combining BLUP and linear programming (Díaz et al., 1999; Famula, 1992).

Furthermore, the restricted aggregate breeding value of Yamada's index (H_R) is given as follows:

$$H_R = \mathbf{a}'\mathbf{g}_R = \mathbf{a}'[\mathbf{I} - \mathbf{G}_0\mathbf{C}(\mathbf{C}'\mathbf{G}_0\mathbf{C})^{-1}\mathbf{C}']\mathbf{g} = \mathbf{a}'\mathbf{k}(\mathbf{k}'\mathbf{G}_0^{-1}\mathbf{k})^{-1}\mathbf{k}'\mathbf{G}_0^{-1}\mathbf{g}.$$

Since $\mathbf{a}'\mathbf{k}$ is a constant, H_R does not depend on the economic weights.

Figure 1 shows a selection for improving trait 1 without changing trait 2, and Figure 2 shows a selection based on relative desired changes. In these figures, the concentric ellipses show the joint probability distribution of the BV for the two traits. The points of BV and RBV are $\mathbf{g}_0$ and $\mathbf{g}_{R0}$, respectively. The RBV of any BV on the plane is the intersection of the vector of the relative desired changes ($\mathbf{g} = \beta\mathbf{k}$) and the tangent ($\mathbf{g}_{R0}'\mathbf{G}_0^{-1}\mathbf{g}_0 = \mathbf{g}_{R0}'\mathbf{G}_0^{-1}\mathbf{g}_{R0}$) of the probability ellipse passing through the BV. Therefore, any BV on the plane is projected onto the point RBV on the vector of the relative desired changes.

Figure 3 shows a restricted selection of three traits that maximizes the aggregate BV, with a change in trait 3 restricted to zero ($r=1$). In this figure, the ellipsoid represents the joint probability distribution of the BV for the three traits. The RBV of any BV in the space is the intersection of the restricted plane ($\mathbf{C}'\mathbf{g} = 0$) and the straight line parallel to $\mathbf{g} = \mathbf{G}_0\mathbf{Ct}$ passing through the BV. The restricted aggregate BV is the inner product of the vector of RBV and the economic weights.

Figure 4 shows the selection based on the relative desired changes for all three traits ($r=2$). Similar to Figures 1 and 2, the RBV of any BV is the intersection of the vector of relative desired changes ($\mathbf{g} = \beta\mathbf{k}$) and the tangent ($\mathbf{g}_{R0}'\mathbf{G}_0^{-1}\mathbf{g}_0 = \mathbf{g}_{R0}'\mathbf{G}_0^{-1}\mathbf{g}_{R0}$) of the probability ellipsoid

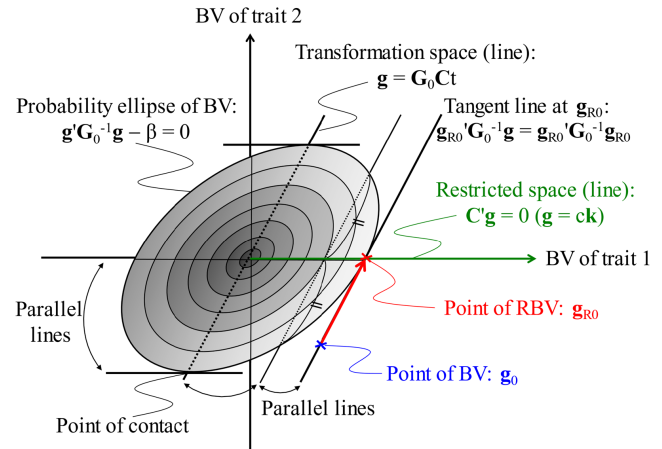


FIGURE 2 The relationship between BV and RBV in a restricted selection based on relative desired changes for two traits. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/jbg.12845)]

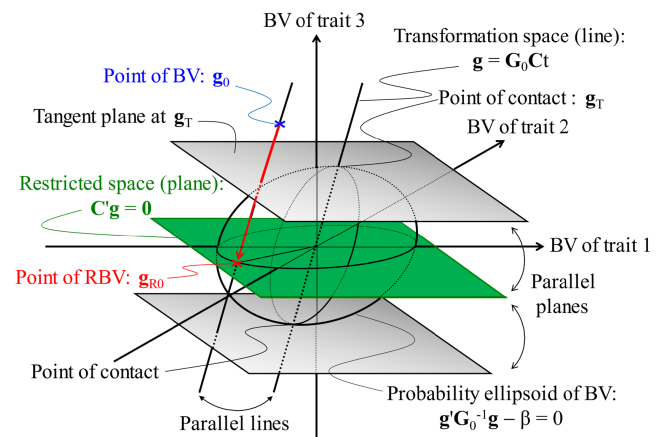


FIGURE 3 The relationship between BV and RBV in a restricted selection of three traits for maximizing the aggregate breeding value with change in trait 3 restricted to zero ($r=1$). [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/jbg.12845)]

passing through the BV. Therefore, the RBV of any BV in the space is projected onto the point RBV on the vector of the relative desired changes.

The geometrical relationship between the BV and RBV can be summarized as follows: Let $\mathbf{C}'\mathbf{g} = 0$ and $\mathbf{g} = \mathbf{G}_0\mathbf{Ct}$ be the restricted space ϕ and transformation space ϕ , respectively. Furthermore, let ω be the space parallel to ϕ . The RBV is the intersection of ϕ and ω , which passes through the BV and is in contact with the ellipsoid. In general, the point BV on the r -dimensional space in restricted selection is projected onto the point RBV on the $(n-r)$ -dimensional space.

3.4 | Numerical example

The relationships between EBV and ERBV and between BV and RBV are illustrated below for a restricted

selection method with three traits. Genetic parameters were the same as those used in the computer simulation (Section 2.3).

Suppose we have data for three animals, as shown in Table 1.

If the relative desired change in traits 1, 2, and 3 based on proportional constraint is 3:4:5, $\mathbf{C}$ is given as follows:

$$\mathbf{C} = \begin{bmatrix} 4 & -3 & 0 \\ 5 & 0 & -3 \end{bmatrix}'$$

From Equations (1) and (2),

$$\mathbf{b}_K = \begin{bmatrix} 0.5101 \\ 0.6492 \\ 0.6956 \end{bmatrix} \text{ and } \mathbf{b}_Y = \begin{bmatrix} 6.4706 \\ 8.2353 \\ 8.8235 \end{bmatrix} = 0.0788 \mathbf{b}_K.$$

Thus, the two different restricted selection indices are equivalent.

From Equation (3), the relationship between EBV ($\hat{\mathbf{g}}_i$) and ERBV ($\hat{\mathbf{g}}_{R_i}$) for the i th animal is as follows:

$$\hat{\mathbf{g}}_{R_i} = [\mathbf{I} - \mathbf{G}'\mathbf{P}^{-1}\mathbf{G}\mathbf{C}(\mathbf{C}'\mathbf{G}'\mathbf{P}^{-1}\mathbf{G}\mathbf{C})^{-1}\mathbf{C}'] \hat{\mathbf{g}}_i = \begin{bmatrix} 0.3599 & 0.2511 & 0.1832 \\ 0.4799 & 0.3348 & 0.2443 \\ 0.5998 & 0.4185 & 0.3053 \end{bmatrix} \hat{\mathbf{g}}_i = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} \left[\begin{array}{ccc} 0.1200 & 0.0837 & 0.0611 \end{array} \right] \hat{\mathbf{g}}_i = \hat{\beta} \mathbf{k}.$$

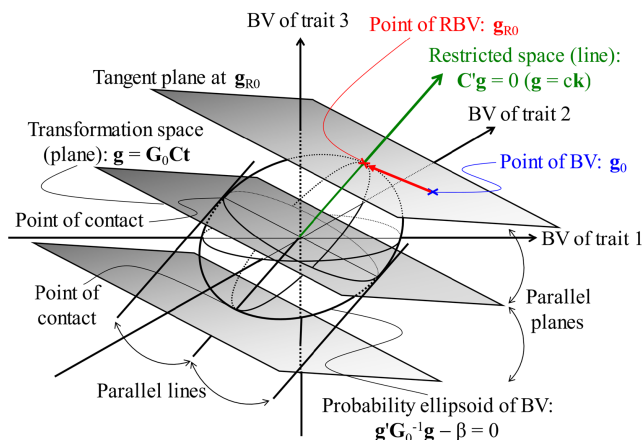


FIGURE 4 The relationship between BV and RBV in a selection based on relative desired changes for all three traits ($r=2$). [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

Animal	Breeding value			Phenotypic value		
	Trait 1	Trait 2	Trait 3	Trait 1	Trait 2	Trait 3
1	0.2	0.2	−0.4	1.2	2.1	4.3
2	−0.1	−0.5	−0.2	1.0	1.6	4.4
3	−0.3	0.3	0.5	0.8	2.2	5.0

As $\hat{\beta} = \begin{bmatrix} 0.1200 & 0.0837 & 0.0611 \end{bmatrix} \hat{\mathbf{g}}_i$ is a scalar, and the ratio of ERBVs was 3:4:5 for all animals. Table 2 shows the EBV, ERBV, and $\hat{\beta}$ for all animals in each selection method.

From Equation (11), the relationship between the true BV ($\mathbf{g}_i$) and RBV ($\mathbf{g}_{R_i}$) for the i th animal is given as follows:

$$\mathbf{g}_{R_i} = \begin{bmatrix} 0.2012 & 0.2561 & 0.2744 \\ 0.2683 & 0.3415 & 0.3659 \\ 0.3354 & 0.4268 & 0.4573 \end{bmatrix} \mathbf{g}_i = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} \left[\begin{array}{ccc} 0.0671 & 0.0854 & 0.0915 \end{array} \right] \mathbf{g}_i = \beta \mathbf{k}.$$

As $\beta = \begin{bmatrix} 0.0671 & 0.0854 & 0.0915 \end{bmatrix} \mathbf{g}_i$ is a scalar, the ratio of RBVs is 3:4:5, which is the same as the ERBVs for all animals. Table 3 shows the BV, RBV, and β for all animals.

3.5 | Computer simulation

From Equation (11), the restricted BVs of the i th individual ($\mathbf{g}_{R_i}$) is a constant multiple of the relative desired changes ($\mathbf{k}$). Let the selection response be $\Delta \mathbf{g}_R$,

$$\Delta \mathbf{g}_R = \frac{\sum \mathbf{g}_{R_i}}{n} = \beta \mathbf{k}.$$

Therefore, $\Delta \mathbf{g}_R$ can be evaluated using the constant β . Table 4 shows the means of responses to selection for the three traits, their evaluation values, and their SE over 100 replicates by computer simulation. In the restricted selection, it is necessary to evaluate not only the magnitude of the response to selection for each trait, but also the ratio of the response to selection. In this example, it is necessary to

TABLE 1 Breeding values and phenotypic values in each of the three traits.

TABLE 2 Estimates of breeding value (EBV), restricted breeding value (ERBV), and selection criterion ($\hat{\beta}$).

Animal	EBV			ERBV			$\hat{\beta}$
	Trait 1	Trait 2	Trait 3	Trait 1	Trait 2	Trait 3	
1	0.498	0.729	1.720	0.677	0.903	1.129	0.226
2	0.444	0.582	1.760	0.629	0.838	1.048	0.210
3	0.462	0.764	2.000	0.725	0.966	1.208	0.241

TABLE 3 True breeding value (BV), restricted breeding value (RBV), and evaluation criterion (β).

Animal	BV			RBV			β
	Trait 1	Trait 2	Trait 3	Trait 1	Trait 2	Trait 3	
1	0.2	0.2	-0.4	-0.018	-0.024	-0.030	-0.006
2	-0.1	0.2	-0.2	-0.203	-0.271	-0.338	-0.068
3	-0.3	-0.5	0.5	0.154	0.205	0.256	0.051

TABLE 4 Response to selection in the three traits and their evaluation values.

			Response to selection ^b							Evaluation value	
Selection method	Genetic parameter ^a	No. dams	Trait 1		Trait 2		Trait 3				
Restricted selection index	T	50	0.210	(0.008)	0.273	(0.009)	0.331	(0.010)	0.070	(0.003)	
		100	0.232	(0.008)	0.314	(0.009)	0.384	(0.010)	0.077	(0.003)	
	F	50	0.149	(0.005)	0.197	(0.009)	0.233	(0.011)	0.050	(0.002)	
		100	0.165	(0.005)	0.218	(0.009)	0.287	(0.011)	0.055	(0.002)	
Restricted BLUP	T	50	0.258	(0.009)	0.336	(0.010)	0.430	(0.011)	0.086	(0.003)	
		100	0.295	(0.009)	0.377	(0.011)	0.473	(0.012)	0.098	(0.003)	
	F	50	0.206	(0.006)	0.278	(0.011)	0.335	(0.013)	0.069	(0.002)	
		100	0.241	(0.006)	0.324	(0.013)	0.397	(0.014)	0.080	(0.002)	

^aT=predicting BVs with true genetic parameters, F=predicting BVs with incorrect genetic parameters.

^bParentheses denote SE.

evaluate that the ratio of response to selection is 3:4:5. The magnitudes of the evaluated values were greater for selection using the restricted BLUP, prediction of restricted BV using true genetic parameters, and larger population size than for the others.

4 | DISCUSSION

Restricted selection indices are the most general linear selection indices; the traditional selection index proposed by Hazel (1943) is a special case of such restricted selection indices (Cerón-Rojas & Crossa, 2020). Kempthorne and Nordskog (1959) derived a selection index that maximizes aggregate BV with changes in some traits restricted to zero. Harville (1975) and Tallis (1985) derived different selection indices to maximize aggregate BV with proportional changes for some traits. However, Itoh and Yamada (1987) proved that the three selection indices are equivalent. In addition to these indices, Yamada et al. (1975) proposed a selection index to achieve the relative desired changes

for all traits. Among the restricted selection indices, the predetermined proportional gain index is more useful than any other selection index because its expected genetic gains can change according to the breeder's interests (Cerón-Rojas et al., 2016).

Kempthorne's selection (zero-change selection) requires economic weights. However, many traits such as body measurement, lifetime production, and disease- and immune-related traits are difficult to calculate economic weights. In addition, because the economy is constantly changing, the economic weights also change over time (Shook, 2006). Therefore, especially in livestock with long generation interval, such as cattle, selection using economic weights does not always result in the maximum economic benefit when offspring are obtained.

As the selection index proposed by Yamada et al. (1975) does not require economic weights for each trait, it has been considered different from selection indices that maximize aggregate BV with constraints (e.g., Kempthorne & Nordskog, 1959) for nearly half a century. However, in this study, I proved that the selection index of Yamada

et al. (1975) is part of Kempthorne and Nordskog (1959). If we have information on pedigree, all these restricted selection indices can be applied to restricted BLUP methods (Quaas & Henderson, 1976; Satoh, 2004). In addition, the selection index method for estimating expected genetic gains proposed by Barwick et al. (2013) can be applied to restricted selection using BV estimated by the BLUP method (Kamo et al., 2023).

I clarified the relationship between EBV using a selection index and ERBV using a restricted selection index. Furthermore, I showed that the ERBVs of the restricted traits were proportional to the relative desired changes in these traits. These are the first findings of the restricted selection index. The latter characteristics have been clarified in restricted BLUP of BVs (Satoh, 1998), but the former relationship, that is, the relationship between BLUP of BVs and restricted BLUP of BVs has not been clarified. Even with restricted BLUP methods, it is necessary to clarify the relationship between EBVs and ERBVs.

When comparing the response to selection for multiple traits, we can use aggregate BV as a criterion (Henderson & Quaas, 1976). However, for restricted selection of desired changes, we need to evaluate the magnitude of genetic improvement in each trait and the deviation from the vector of desired changes at the same time. However, to date, there is no established criterion to compare the response to selection with constraints. Satoh (2019) derived the relationship between BV and RBV by limit, as the error tends to 0 (heritability close to 1) in a restricted mixed model. In this study, the relationship between BV and RBV was derived using the multivariate normal distribution of BV and the $(n - r)$ -dimensional restricted space.

For all constraints, any BVs in each animal with constraints were linearly transformed to RBVs by the transformation matrix $\mathbf{I} - \mathbf{G}_0 \mathbf{C} (\mathbf{C}' \mathbf{G}_0 \mathbf{C})^{-1} \mathbf{C}'$, with the vector space of the RBV represented in the $(n - r)$ dimension, where n is the dimension of the BV and r is the rank of the restriction matrix $\mathbf{C}$. Rank r is equal to the number of traits for no-change constraints or the number of traits for proportional changes minus one. If constraints include both no changes and proportional changes, r is the number of traits for no-change constraints and for proportional changes minus 1. For example, for 2 traits, if the restricted trait is one, the RBVs are straight lines; for three traits, if the restricted trait is one, the RBVs are flat; and if the restricted trait is two, the RBVs are straight lines. The geometrical relationship between RBV and BV is applicable to four or more traits. If $n = 4$ and $r = 1$, the restricted and transformation spaces are three- and one-dimensional vector spaces, respectively. Unfortunately, it is difficult to represent this in visual form.

If constraints with changes in some traits are restricted to zero, no economic weights corresponding to the restricted traits are needed. Additionally, if $n - r = 1$, the restricted space is one-dimensional, that is, a straight line, in which case no economic weights for any traits are needed. Otherwise, economic weights are needed for all traits. Therefore, if $n - r > 1$, the inner product $(\mathbf{a}' \hat{\mathbf{g}}_R)$ of the vector of economic weights and the vector of ERBV is the criterion for selection. However, because $\mathbf{a}' \hat{\mathbf{g}}_R$ is always smaller than $\mathbf{a}' \mathbf{g}$, we cannot evaluate the response to restricted selection using $\mathbf{a}' \hat{\mathbf{g}}_R$ in a simulation study. In addition, if $n - r = 1$, the evaluation of the response to restricted selection is difficult, because there are many criteria for evaluating the response to selection. For example, Toro and Silió (1992), Díaz et al. (1999), and Ieiri et al. (2004) used the means and variances of the achieved genetic response to selection for each of the two traits as the criteria for comparing restricted selection strategies in stochastic simulation studies. Furthermore, if restricted selection for more than two traits is conducted in a simulation study, comparison is more difficult. However, Equation (11) can be used as the sole criterion for evaluating the response to multi-trait selection with constraints when carrying out simulation studies using not only restricted selection indices, but also restricted BLUP.

Equation (11) or β can be used not only when $\mathbf{g}$ is known in advance, as in computer simulations, but also in other cases. If the population size (n) is large, the vector of population means of the phenotypic values ($\mathbf{p}$) after selection is given as follows:

$$\lim_{n \rightarrow \max} \frac{\sum_{i=1}^n \hat{\mathbf{g}}_R}{n} = \lim_{n \rightarrow \max} \frac{\mathbf{k}' \mathbf{G}^{-1} \sum_{i=1}^n \mathbf{p}_i}{n} \cong \left[\mathbf{I} - \mathbf{G}_0 \mathbf{C} (\mathbf{C}' \mathbf{G}_0 \mathbf{C})^{-1} \mathbf{C}' \right] \bar{\mathbf{g}} = \beta \mathbf{k},$$

where $\bar{\mathbf{g}}$ is the vector of BV means in the population. In the case of the restricted BLUP method, restricted BLUP of BVs can be used instead of $\hat{\mathbf{g}}_R$ in the above equation to evaluate the genetic performance of the population by the single criterion, β . In other words, when the breeding goal is the predetermined desired changes, the genetic performance of the population after selection can be evaluated by β if the phenotypic or restricted EBV means are available for each trait. By using β , it is also possible to compare the genetic gains obtained by different selection.

Yamada's selection (selection for desired changes) has been cited in many publications recently, for example, Slagboom et al. (2018) for dairy cow simulation, Dzama et al. (2001) for beef cattle simulation, Suzuki et al. (2005), Kadowaki et al. (2012), and Ohnishi and Satoh (2018) for pig selection experiments, Noda et al. (2002) and Ochiai

et al. (2005) for chicken experiments, and Lwelamira et al. (2008) for chicken simulation. There are also many studies that have predicted genetic gains by selection for desired changes, for example, reproductive and production traits (Kaushik & Khanna, 2003), lactation persistence (Lin & Togashi, 2005; Togashi & Lin, 2003) in dairy cows, red blood cell volume and weight gain in beef cattle (Fukasawa et al., 2002), rabbit carcass traits (Ács et al., 2019), and quail weight and egg production (Hidalgo et al., 2014). Although there are few examples of Kempthorne's selection (zero-restricted selection), there are, for example, studies of body weight and height in dogs (Helmink et al., 2003). However, the zero-restricted selection may be used because the selected traits have reached optimum values for some traits. For example, if egg weight, flesh colour, flesh pH, etc., are improved to optimum values, zero-restricted selection will become necessary. Although zero-restricted selection may not be used very often at present, the number of traits that will be subject to zero-restricted selection will increase in the near future.

5 | CONCLUSIONS

First, I proved that the restricted selection index that achieves the relative desired changes for all traits (Yamada et al., 1975) is part of the restricted selection index that maximizes the aggregate BV with changes in some traits restricted to zero (Kempthorne & Nordskog, 1959). Second, I clarified that (1) the relationship between the BVs estimated by the ordinary selection index and those estimated by the restricted selection index (ERBVs) and (2) the ERBVs of the restricted traits are proportional to the relative desired changes. Third, I geometrically represented the concept of the relationship between BV and restricted BV. This result can be used as the sole criterion for evaluating responses to selection with constraints.

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CONFLICT OF INTEREST STATEMENT

The author declares no conflict of interest.

DATA AVAILABILITY STATEMENT

All relevant data are included in the manuscript and its associated files.

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APPENDIX 1

The Proof of Lemma 1 is as follows:

As $\mathbf{V}$ is symmetric and a positive definite matrix, $\mathbf{V} = \mathbf{T}\mathbf{T}'$ for $\mathbf{T}_{n \times n}$ non-singular. Now consider

$$\mathbf{M} = \mathbf{I} - \mathbf{T}^{-1}\mathbf{X}(\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'(\mathbf{T}')^{-1} - \mathbf{T}'\mathbf{K}(\mathbf{K}'\mathbf{V}\mathbf{K})^{-1}\mathbf{K}'\mathbf{T}.$$

From $\mathbf{M} = \mathbf{M}\mathbf{M} = \mathbf{M}'\mathbf{M}$,

$$tr(\mathbf{M}'\mathbf{M}) = tr(\mathbf{M}) = tr(\mathbf{I}) - tr\left[\mathbf{T}^{-1}\mathbf{X}(\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'(\mathbf{T}')^{-1}\right] - tr\left[\mathbf{T}'\mathbf{K}(\mathbf{K}'\mathbf{V}\mathbf{K})^{-1}\mathbf{K}'\mathbf{T}\right] = n - r - (n - r) = 0.$$

As $tr(\mathbf{M}'\mathbf{M}) = 0$, then $\mathbf{M} = \mathbf{0}$. Therefore,

$$\mathbf{T}'\mathbf{K}(\mathbf{K}'\mathbf{V}\mathbf{K})^{-1}\mathbf{K}'\mathbf{T} = \mathbf{I} - \mathbf{T}^{-1}\mathbf{X}(\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'(\mathbf{T}')^{-1}.$$

Pre-multiplication by $(\mathbf{T}')^{-1}$ and post-multiplication by $\mathbf{T}^{-1}$ yields Lemma 1.

For another proof, see Itoh and Yamada (1987).

APPENDIX 2

From Lemma 1,

$$\begin{aligned}\hat{\mathbf{g}}_R &= \mathbf{G}'\mathbf{P}^{-1}\left[\mathbf{I} - \mathbf{G}\mathbf{C}(\mathbf{C}'\mathbf{G}'\mathbf{P}^{-1}\mathbf{G}\mathbf{C})^{-1}\mathbf{C}'\mathbf{G}'\mathbf{P}^{-1}\right]\mathbf{p} \\ &= \begin{bmatrix} \mathbf{G}'_1 - \mathbf{G}'_1\mathbf{P}^{-1}\mathbf{G}_1\mathbf{C}_0(\mathbf{C}'_0\mathbf{G}'_1\mathbf{P}^{-1}\mathbf{G}_1\mathbf{C}_0)^{-1}\mathbf{C}'_0\mathbf{G}'_1 \\ \mathbf{G}'_2 - \mathbf{G}'_2\mathbf{P}^{-1}\mathbf{G}_1\mathbf{C}_0(\mathbf{C}'_0\mathbf{G}'_1\mathbf{P}^{-1}\mathbf{G}_1\mathbf{C}_0)^{-1}\mathbf{C}'_0\mathbf{G}'_1 \end{bmatrix} \mathbf{P}^{-1}\mathbf{p} \\ &= \begin{bmatrix} \left[\mathbf{I} - \mathbf{G}'_1\mathbf{P}^{-1}\mathbf{G}_1\mathbf{C}_0(\mathbf{C}'_0\mathbf{G}'_1\mathbf{P}^{-1}\mathbf{G}_1\mathbf{C}_0)^{-1}\mathbf{C}'_0\right]\mathbf{G}'_1\mathbf{P}^{-1}\mathbf{G}_1(\mathbf{G}'_1\mathbf{P}^{-1}\mathbf{G}_1)^{-1}\mathbf{G}'_1 \\ \mathbf{G}'_2 - \mathbf{P}^{-1}\mathbf{G}_1\mathbf{C}_0(\mathbf{C}'_0\mathbf{G}'_1\mathbf{P}^{-1}\mathbf{G}_1\mathbf{C}_0)^{-1}\mathbf{C}'_0\mathbf{G}'_1 \end{bmatrix} \mathbf{P}^{-1}\mathbf{p} \\ \hat{\mathbf{g}}_R &= \begin{bmatrix} \mathbf{k}_0\left[\mathbf{k}'_0(\mathbf{G}'_1\mathbf{P}^{-1}\mathbf{G}_1)^{-1}\mathbf{k}_0\right]^{-1}\mathbf{k}'_0(\mathbf{G}'_1\mathbf{P}^{-1}\mathbf{G}_1)^{-1}\mathbf{G}'_1 \\ \mathbf{G}'_2 - \mathbf{P}^{-1}\mathbf{G}_1\mathbf{C}_0(\mathbf{C}'_0\mathbf{G}'_1\mathbf{P}^{-1}\mathbf{G}_1\mathbf{C}_0)^{-1}\mathbf{C}'_0\mathbf{G}'_1 \end{bmatrix} \mathbf{P}^{-1}\mathbf{p}\end{aligned}$$

where $\mathbf{k}'_0(\mathbf{G}'_1\mathbf{P}^{-1}\mathbf{G}_1)^{-1}\mathbf{k}_0$ is a constant and the magnitude $\mathbf{k}'_0(\mathbf{G}'_1\mathbf{P}^{-1}\mathbf{G}_1)^{-1}\mathbf{G}'_1\mathbf{P}^{-1}\mathbf{p}$ is a scalar that depends on $\mathbf{p}$. Then, let

$$\hat{\beta}_0 = \left[\mathbf{k}'_0(\mathbf{G}'_1\mathbf{P}^{-1}\mathbf{G}_1)^{-1}\mathbf{k}_0\right]^{-1}\mathbf{k}'_0(\mathbf{G}'_1\mathbf{P}^{-1}\mathbf{G}_1)^{-1}\mathbf{G}'_1\mathbf{P}^{-1}\mathbf{p} \text{ be } \hat{\mathbf{g}}_R = \begin{bmatrix} \hat{\beta}_0\mathbf{k}_0 \\ \left[\mathbf{G}'_2 - \mathbf{P}^{-1}\mathbf{G}_1\mathbf{C}_0(\mathbf{C}'_0\mathbf{G}'_1\mathbf{P}^{-1}\mathbf{G}_1\mathbf{C}_0)^{-1}\mathbf{C}'_0\mathbf{G}'_1\right]\mathbf{P}^{-1}\mathbf{p} \end{bmatrix}.$$